

Model Comparison

Objective

The purpose of this analysis is to compare simple and multiple linear regression models to determine which model best explains the variation in **Monthly Sales** and provides the most useful business insights.

Model 1 – Simple Linear Regression

Model Name

Simple Regression Model 1

Dependent Variable

Monthly Sales

Independent Variable

Marketing Spend

Regression Equation

Monthly Sales = $\beta_0 + \beta_1(\text{Marketing Spend})$

Results

- R-squared: ____
- Coefficient (Marketing Spend): ____
- P-value: ____

Business Interpretation

This model evaluates whether increases in marketing spending are associated with higher monthly sales.

If the coefficient is positive and statistically significant (p-value < 0.05), increasing marketing investment is associated with higher sales performance.

Business Usefulness

- Easy to interpret.
- Useful for understanding the isolated effect of marketing investment.
- Does not account for other business factors affecting sales.

Limitations

- Ignores additional variables that may influence sales.
 - Lower explanatory power compared to multiple regression.
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Model 2 – Simple Linear Regression

Model Name

Simple Regression Model 2

Dependent Variable

Monthly Sales

Independent Variable

Footfall

Regression Equation

Monthly Sales = $\beta_0 + \beta_1(\text{Footfall})$

Results

- R-squared: ____
- Coefficient (Footfall): ____
- P-value: ____

Business Interpretation

This model measures the relationship between customer footfall and monthly sales.

A positive and statistically significant coefficient suggests that higher customer traffic is associated with increased sales.

Business Usefulness

- Demonstrates the relationship between customer visits and revenue.
- Easy for management to interpret.

Limitations

- Does not include marketing, inventory, pricing, or store characteristics.
 - Cannot explain all variation in monthly sales.
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Model 3 – Multiple Linear Regression

Model Name

Multiple Regression Model

Dependent Variable

Monthly Sales

Independent Variables

- Marketing Spend
- Footfall
- Inventory Availability Percentage
- Region Dummy Variable (or Store Type Dummy Variable)

Regression Equation

Monthly Sales = $\beta_0 + \beta_1(\text{Marketing Spend}) + \beta_2(\text{Footfall}) + \beta_3(\text{Inventory Availability}) + \beta_4(\text{Dummy Variable})$

Results

- R-squared: ____
- Adjusted R-squared: ____

Significant Variables

Variable	Coefficient	P-value	Significant (Yes/No)
Marketing Spend	---	---	---
Footfall	---	---	---
Inventory Availability	---	---	---
Dummy Variable	---	---	---

Business Interpretation

The multiple regression model evaluates the combined effect of several business factors on monthly sales.

Variables with statistically significant p-values provide the strongest evidence of association with sales performance.

Variables with large coefficients indicate stronger estimated impacts while holding the other predictors constant.

Business Usefulness

- Explains more variation in monthly sales than individual models.

- Supports strategic business decisions.
- Accounts for multiple business drivers simultaneously.
- Better suited for forecasting than simple regression.

Limitations

- Regression measures association rather than causation.
- Model performance depends on data quality.
- Important external variables may not be included.
- Multicollinearity between predictors can affect coefficient estimates.

Model Comparison Summary

Model	Variables Used	R-squared	Significant Variables	Business Usefulness	Limitations
Simple Regression 1	Marketing Spend	—	Marketing Spend	Measures individual effect of marketing	Ignores other variables
Simple Regression 2	Footfall	—	Footfall	Measures relationship between traffic and sales	Limited explanatory power
Multiple Regression	Marketing Spend, Footfall, Inventory Availability, Dummy Variable	—	Significant predictors based on p-values	Best overall model for decision-making	Association only; not proof of causation

Final Model Selection

The **Multiple Regression Model** is selected as the preferred model because it:

- Includes multiple business drivers simultaneously.
- Typically provides the highest explanatory power (R-squared).
- Controls for the influence of other variables.
- Produces more realistic business insights.
- Supports better forecasting and decision-making.

Overall Conclusion

The comparison indicates that while simple regression models are useful for understanding individual relationships, the multiple regression model provides a more comprehensive explanation of monthly sales performance.

Leadership should base strategic decisions primarily on the multiple regression model while considering statistical significance, business context, and practical limitations. Regression identifies associations between variables but should not be interpreted as definitive evidence of causation.